MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATIONS, January 2026
Fourth Semester
MODEL QUESTION
DSE – MG4DSEZGY200
GENERAL TOXICOLOGY

Duration: 2hr Maximum Mark:70

PART A – MCQ (1 mark). Answer all (10 X 1 =10 marks)

1. LD_{50} refers to the dose that causes death in……..of test animals. [U] [CO:1]
a) 25% b) 50% c) 75% d) 100%

2. Tartrazine is used as a …….. [U] [CO:2]
a) Preservative b) Colouring agent c) Antioxidant d)Sweetener

3. Chronic exposure to Aflatoxins primarily causes ……. [U] [CO: 2]
a) Kidney Failure b) Liver cancer c) Anemia d) Lung Fibrosis

4. The term "Xenobiotics" is derived from the Greek word meaning…… [A] [CO: 3]
a) Foreign and animal b) Life and chemical
c) Foreign and life d) Chemical and toxic

5. Lipid soluble xenobiotics tend to accumulate primarily in …… [A] [CO: 3]
a) Bone b) Kidney c) Fat tissue d) Blood plasma

6. The toxic effect of methyl mercury in fish is an example of……. [A] [CO: 3]
a) Biodegradation b) Biotransformation
c) Detoxification d) Biological magnification

7. Anaphylaxis in cosmetic exposure is most often triggered by ……. [U] [CO: 4]
a) SLS b) Parabens c) PPD OR Persulfates d) Dimethicone

8. Which Pollutant accumulates in marine mammal blubber? [C] [CO: 4]
a) Ammonia b) Sulfur dioxide c) Organophosphates d) Persistent Organic Pollutants

9. Fungus that pose health risks in human beings…….. [U] [CO: 5]
a) *Aspergillus* b) *Giardia* c) *Salmonella* d) *Trypanosoma*

10. A major challenge in India's occupational health system is …… [A] [CO5]
a) Excessive safety compliance b) Over reporting accidents
c) Poor enforcement of regulations d) No informal sector involvement

PART B – Short Answer (3 marks) Write any 10. (10 X3 = 30 marks)

11. Mention the 3Rs principle and its importance in animal research. [U] [CO: 1]

12. Mention any two international agencies that regulate food additive safety. [U] [CO:2]

13. Distinguish between Phase I and Phase II reactions [A] [CO: 3]

14. Describe the toxic effect of DDT on birds. [A] [CO: 3]

15. What are the key symptoms and causes of Itai-Itai disease and Minimata disease? [A] [CO:3]

16. List any four toxic components commonly found in shampoos. [U] [CO: 4]

17. What are the symptoms of occupational asthma in hairdressers? [U] [CO: 4]

18. Explain how amphibian skin structure contributes to chemical vulnerability. [U] [CO: 4]

19. Mention the physiological effects of opioids and stimulants. [U] [CO: 4]

20. Indicate the role of ergonomics in human capacities and limitations. [A] [CO: 5]

21. What are corrosive chemicals? Give two examples. [U] [CO: 5]

22. What are the key elements emphasized by ISO45001? [A] [CO: 5]

PART C – Short Essay (5 marks) Write any 6 (4 X5 =20 marks)

23. Write an account on ethical considerations, CPCSEA regulations and OECD guidelines in animal toxicology. [U] [CO:1]

24. Explain the mechanism, symptoms and prevention of botulinum poisoning. [U] [CO: 2]

25. Explain the different routes and modes of excretion of Xenobiotics from the blood. [A] [CO: 3]

26. Describe how specialized barriers such as blood-brain barrier, protect vital organs from toxicants.[A] [CO:3]

27. Describe the allergic contact dermatitis and its association with cosmetic chemicals. [U] [CO: 4]

28. Describe the fundamental rights under Indian Constitution. [U] [CO: 4]

29. Describe biological hazards including the modes of transmission of pathogens. [A] [CO: 5]

30. Discuss engineering measures used in controlling occupational hazards and explain why they are highly effective. [A] [CO: 5]

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